DONOR RETENTION & LAPSE ANALYSIS

Who stops giving, who is worth winning back, and when giving peaks

Python, PostgreSQL, machine learning and Salesforce

$1.62M	5,000	8,816	1,405
total giving, 2023 to 2025	donors	gifts	lapsed donors analyzed

Prepared by	Imane Khtib
Data	Simulated donor and gift records (seed 42)
Tools	Python (pandas, NumPy, matplotlib, seaborn, scikit-learn), Jupyter, PostgreSQL, Salesforce

Simulated data, for learning and portfolio use

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1. Executive Summary

The headline

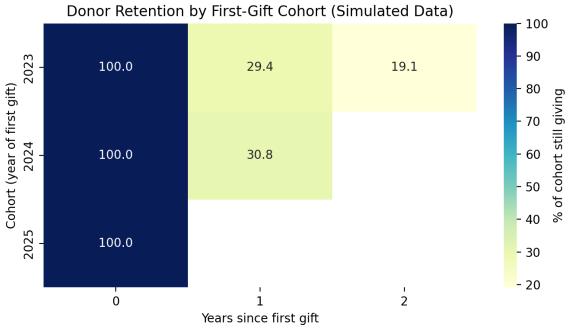
Most new donors never give a second time, and a small group holds most of the recoverable value.

Nonprofits lose most first-time donors and depend on a small group of loyal and major givers. This project analyzes 8,816 gifts from 5,000 simulated donors (2023 to 2025, $1,616,405) to answer four questions: how many donors come back, who stops giving, which lapsed donors are worth contacting first, and when and through which campaigns giving arrives. A predictive model then ranks donors by their risk of lapsing.

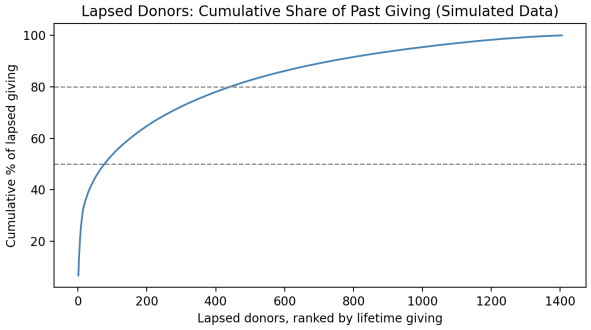
~30%	77 of 1,405	35%	0.82
of new donors give again	lapsed donors hold half of lapsed giving	of annual giving arrives in Nov and Dec	ROC-AUC of the lapse-risk model

Key results

Finding	Result	What to do
First-year retention	About 30% of new donors give again (29.4% for the 2023 cohort, 30.8% for 2024)	Send a thank-you and a second ask soon after the first gift
Survivors give more	The 2023 cohort's average gift rises from $141 to $241 to $267	Steward loyal and major donors personally
Growth depends on new donors	Each older cohort shrinks every year, and 62% of 2025 giving came from returning donors	Track retention by cohort, not only total giving
Lapsed value is concentrated	77 of 1,405 lapsed donors (about 5%) hold half of lapsed giving	Personal outreach to those 77, a lighter touch for the next few hundred
Giving is seasonal	November and December bring in about 35% of annual giving	Plan the Year-End Appeal early
Lapse is predictable	ROC-AUC 0.82. The highest-risk 30% of donors lapse at about 94%, the lowest-risk 10% at 16%	Use risk scores to rank win-back outreach



Cohort retention: about 3 in 10 new donors return



Lapsed giving: a few donors hold most of the value

Recommendations

1. **Improve the first renewal:** thank-you within days, second ask within a few months.
2. **Steward loyal and major donors:** personal contact for the donors who stay.
3. **Tier the win-back effort:** personal contact for the top 77 lapsed donors, lighter touch for the next few hundred.
4. **Rank outreach by risk and value:** combine each donor's lapse-risk score with past giving.
5. **Plan the Year-End Appeal early:** it is the strongest named campaign ($473,775).
6. **Keep acquiring, and track cohorts:** growth should not hide a weakening donor base.

Important note: all data is simulated. Results show how the method works and what it can reveal. They do not describe a real organization.

2. Business Problem and Questions

Fundraising teams lose most first-time donors and rely on a small number of loyal and major donors. To spend limited staff time and budget well, a team needs to know who is leaving, who is worth winning back, and when giving peaks. This project builds that analysis from the ground up.

Question	Method	Where answered
What share of donors come back each year?	Retention by year and by cohort (SQL, Python heatmap)	Section 5.1
Do returning donors give differently?	Giving and average gift by cohort and year (SQL)	Section 5.2
Who has lapsed, and who matters most?	Lapsed list ranked by lifetime giving, with cumulative share (SQL)	Section 5.3
Which campaigns raise the most, and when does giving happen?	Total, average and share by campaign; gifts and dollars by month (SQL, Python)	Section 5.4
Does gift size differ by donor type or channel?	Average gift by channel and donor type (Python)	Section 5.5
Who is likely to lapse next?	Logistic regression and random forest (Python)	Section 6

3. The Data

3.1 Why simulated data

Real donor records contain personal information and are rarely available for practice. A simulation avoids privacy issues and has a second benefit: because the rules that generate the data are known, the analysis can be checked against them. If the analysis recovers the patterns that were built in, the method works.

3.2 Database tables

donors has one row per donor (5,000 rows).

Column	Type	Description
donor_id	VARCHAR(10), primary key	Unique ID such as D00001
donor_type	VARCHAR(20)	Individual (92%) or Corporate (8%)
state	CHAR(2)	CT 50%, RI 20%, MA 15%, NY 15%

gifts has one row per gift (8,816 rows).

Column	Type	Description
gift_id	VARCHAR(10), primary key	Unique ID such as G000001
donor_id	VARCHAR(10), foreign key	Links each gift to a donor in the donors table
gift_date	DATE	Date of the gift, 2023 to 2025
amount	NUMERIC(10,2)	Gift size in dollars
campaign	VARCHAR(30)	General Fund, Year-End Appeal, Spring Drive or Disaster Relief
channel	VARCHAR(20)	Online 55%, Mail 25%, Event 15%, Phone 5%

3.3 How the gifts were simulated

Each donor was secretly assigned a giver segment that controls how often they renew and how much they give. The segment is not stored in the tables, so the analysis has to discover the patterns on its own.

Segment	Share of donors	Donors	Chance of giving again	Typical gift (median)
One-time	55%	2,722	5%	$35
Occasional	30%	1,555	45%	$60
Loyal	12%	587	85%	$120
Major	3%	136	90%	$1,500

- **First-gift year:** 40% start in 2023, 33% in 2024, 27% in 2025.
- **Reactivation:** a donor who skipped a year can return, at 25% of their normal renewal chance.
- **Gift size:** drawn from a lognormal distribution, so most gifts are small and a few are very large. Corporate gifts are multiplied by 3.
- **Seasonality:** December carries 23% of gifts and November 12%, with 5% to 8% for most other months.
- **Campaigns:** Disaster Relief in September 2023, October 2024 and August 2025 (70% of gifts those months). Year-End Appeal in November and December (80%). Spring Drive from March to May (60%). Everything else is General Fund.
- **Reproducibility:** the random seed is fixed at 42, so the same data is produced each time.

3.4 Resulting dataset

Year	Gifts	Total given	Donors giving
2023	2,523	$356,625	1,971
2024	3,048	$554,545	2,230
2025	3,245	$705,235	2,264
Total	**8,816**	**$1,616,405**	**5,000 unique**

The average gift across all years is about $183.

4. Methodology

4.1 Workflow

1. **Simulate** the donors and gifts in Python (pandas, NumPy).
2. **Load** both tables into PostgreSQL with primary and foreign keys.
3. **Validate** the data with SQL quality checks.
4. **Analyze** retention, cohorts, lapsed donors and campaigns with SQL.
5. **Visualize** the results in Python (matplotlib, seaborn).
6. **Model** lapse risk in Python (scikit-learn).
7. **Extend** to a CRM by loading a sample into Salesforce and building reports and a dashboard.

4.2 Key definitions

Term	Definition used in this project
Retention rate	Share of last year's donors who gave again this year
Cohort	A group of donors who made their first gift in the same year
Years since first gift	Giving year minus cohort year. Year 0 is the first-gift year, so it is always 100%
Lapsed donor	A donor who gave in 2024 but made no gift in 2025
Lifetime giving	Total of all gifts a donor made across all years

4.3 Data quality validation

Before any analysis, six SQL checks confirmed the data was complete and consistent. All six passed, and the row counts match the Python output exactly.

Check	What it catches	Result
donors rows	Missing or extra donors after import	5,000, pass
gifts rows	Missing or extra gifts after import	8,816, pass
null amounts or dates	Blank values that break totals and date logic	0, pass
amounts <= 0	Impossible gift values	0, pass
duplicate gift IDs	Gifts loaded twice	0, pass
gifts with no matching donor	Orphan gifts (broken foreign key)	0, pass

4.4 SQL techniques

- **CTEs (WITH clauses)** build each analysis in readable, named steps.
- **SELECT DISTINCT** reduces gifts to one row per donor per year, so a donor with three gifts is counted once.
- **Self-join with LEFT JOIN** matches each donor-year to the same donor's next year to find returning donors.
- **Anti-join** (LEFT JOIN where the match is NULL) finds donors who gave in 2024 but not in 2025.
- **Window functions** (RANK, ROW_NUMBER, running SUM over total SUM) order lapsed donors and compute cumulative share.
- **FILTER with MIN** finds the first donor number at which the cumulative share reaches 50% and 80%.

5. Findings

5.1 Retention: only 3 in 10 new donors give again

Retention was measured two ways. The cohort view follows each group of new donors over time. The all-donor view asks what share of last year's donors returned, including long-time donors.

Cohort	Donors	Year 0	Year 1	Year 2
2023	1,971	100.0%	29.4% (579)	19.1% (377)
2024	1,651	100.0%	30.8% (509)	not yet
2025	1,378	100.0%	not yet	not yet

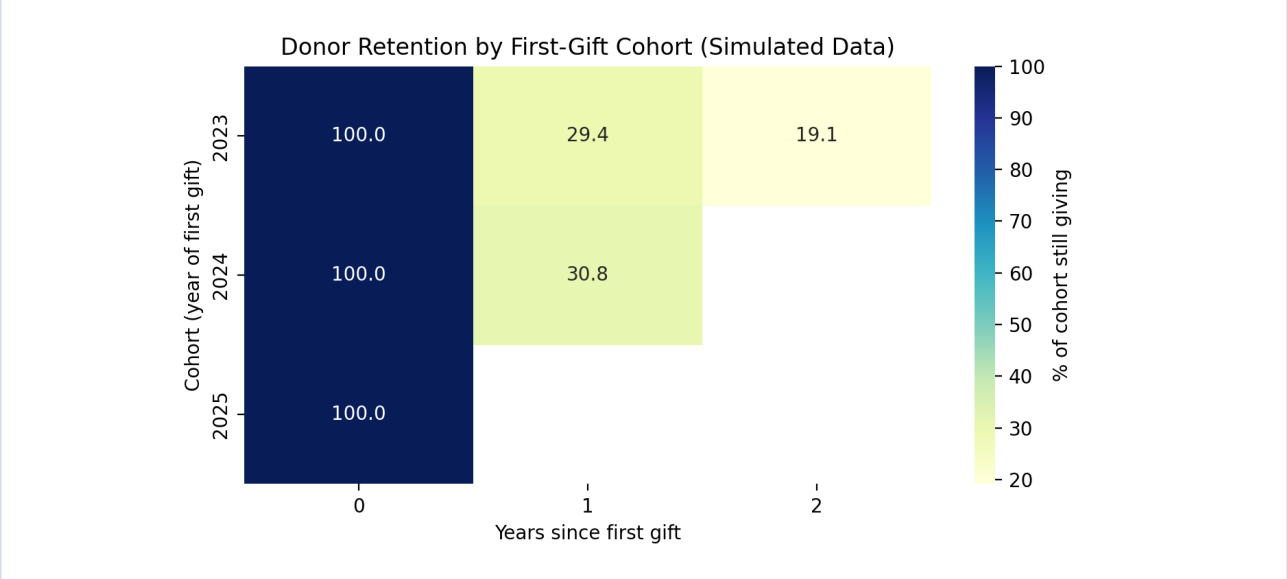


Figure 1. Cohort retention heatmap (each cell is the percent of a cohort still giving)

What this means

About 30% of new donors give a second time. The loss in year 2 is smaller (29.4% falls to 19.1%), so donors who survive the first year tend to stay. The simulation settings predict a blended first-year retention of about 29.2%, and the analysis found 29.4% and 30.8%, so the method recovered the pattern that was built in.
So what: the biggest loss happens right after the first gift, so that is where retention work pays off most.

Why the all-donor view differs. Across all donors, retention was 29.4% from 2023 to 2024 and 37.0% from 2024 to 2025. The second rate is higher because by 2024 the donor base includes loyal returning donors as well as new ones. (The SQL output shows 0.0% for 2025 only because there is no 2026 data yet, so that row is ignored.)

5.2 Revenue by cohort: growth relies on new donors

Giving year	2023 cohort	2024 cohort	2025 cohort	Total
2023	$356,625			$356,625
2024	$216,895	$337,650		$554,545
2025	$171,580	$266,350	$267,305	$705,235

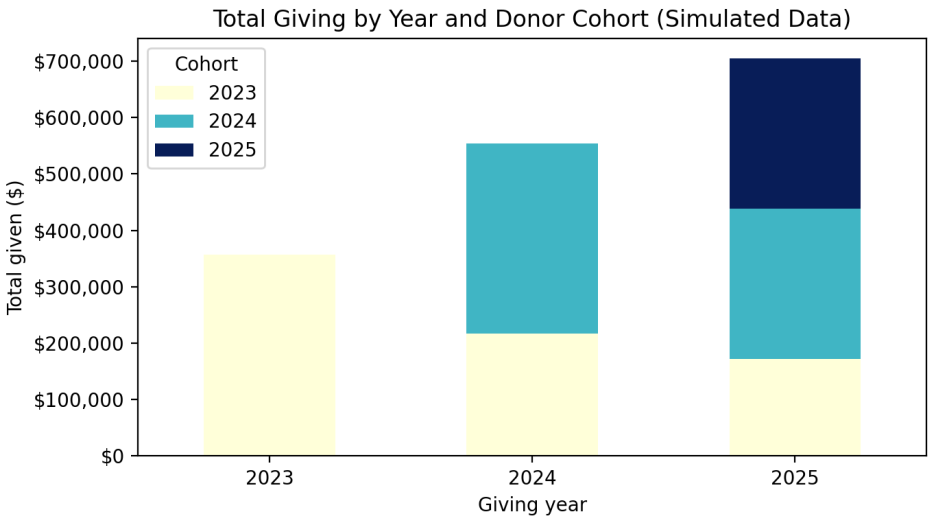


Figure 2. Total giving by year, stacked by donor cohort

Total giving grew 55.5% from 2023 to 2024 and 27.2% from 2024 to 2025. Each older cohort shrinks every year: the 2023 cohort gives $356,625, then $216,895 (down 39%), then $171,580 (down 21%). **Survivors give more.** The SQL cohort query shows average gift rising with donor age:

Cohort	Giving year	Gifts	Donors	Total giving	Average gift
2023	2023	2,523	1,971	$356,625	$141.35
2023	2024	898	579	$216,895	$241.53
2023	2025	642	377	$171,580	$267.26
2024	2024	2,150	1,651	$337,650	$157.05
2024	2025	839	509	$266,350	$317.46
2025	2025	1,764	1,378	$267,305	$151.53

What this means

In 2025, returning donors from the 2023 and 2024 cohorts gave $437,930 (62% of the year) and brand-new donors gave $267,305 (38%). Average gifts jump after year 1 because the donors who stay are mostly loyal and major givers. This is a survivor effect: fewer people, but bigger gifts.

So what: if new donor acquisition slowed, total giving would fall quickly. Retention should be reported by cohort so growth does not hide a weakening base.

5.3 Lapsed donors: a small group holds most of the value

There are 1,405 lapsed donors (gave in 2024, did not give in 2025) with $212,325 in lifetime giving. Ranking them by lifetime giving shows how concentrated that value is.

Share of lapsed giving	Donors needed	Share of lapsed donors
50%	77	about 5%
80%	441	about 31%
100%	1,405	100%

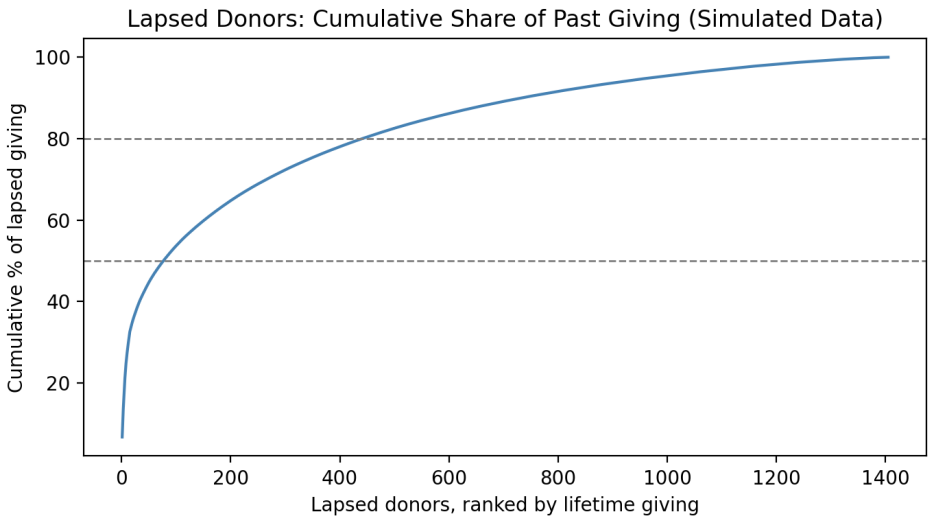


Figure 3. Cumulative share of lapsed giving, donors ranked by lifetime giving

What this means

A win-back effort should not treat all 1,405 donors alike. Personal calls or letters to the top 77 are the best use of staff time. A lighter, cheaper touch such as email or a mailer fits the next few hundred, and little should be spent on the long tail, which is mostly small one-time donors.

5.4 Campaigns and seasonality

Campaign	Gifts	Donors	Total giving	Average gift	Share
General Fund	4,621	3,269	$866,315	$187.47	53.6%
Year-End Appeal	2,537	2,024	$473,775	$186.75	29.3%
Spring Drive	1,259	1,115	$182,260	$144.77	11.3%
Disaster Relief	399	389	$94,055	$235.73	5.8%

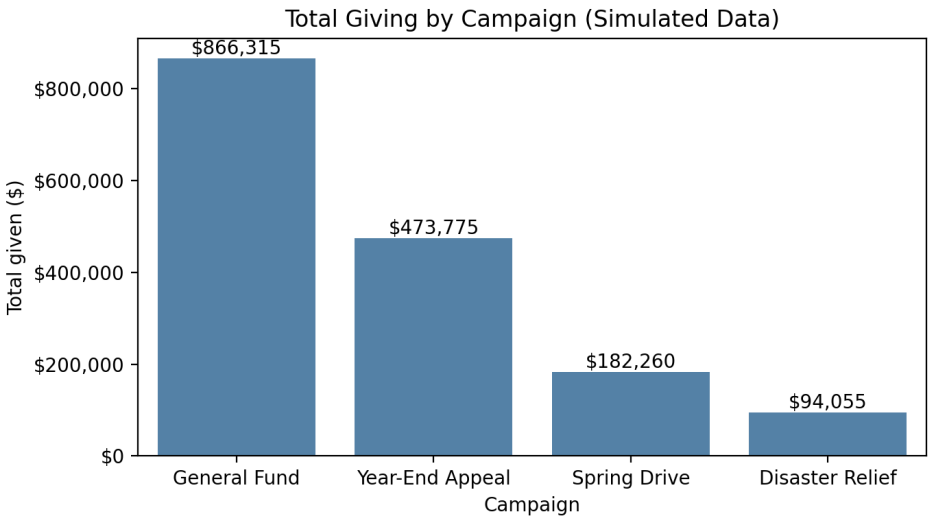


Figure 4. Total giving by campaign

General Fund is the largest because it is the catch-all label for every gift that did not fall into a named campaign, so its size reflects volume, not performance. Among named campaigns, Year-End Appeal raises nearly half a million dollars in two months. Disaster Relief is small in total, but its average gift is the highest at $235.73. The same donor can give to more than one campaign, which is why donor counts add up to more than 5,000.

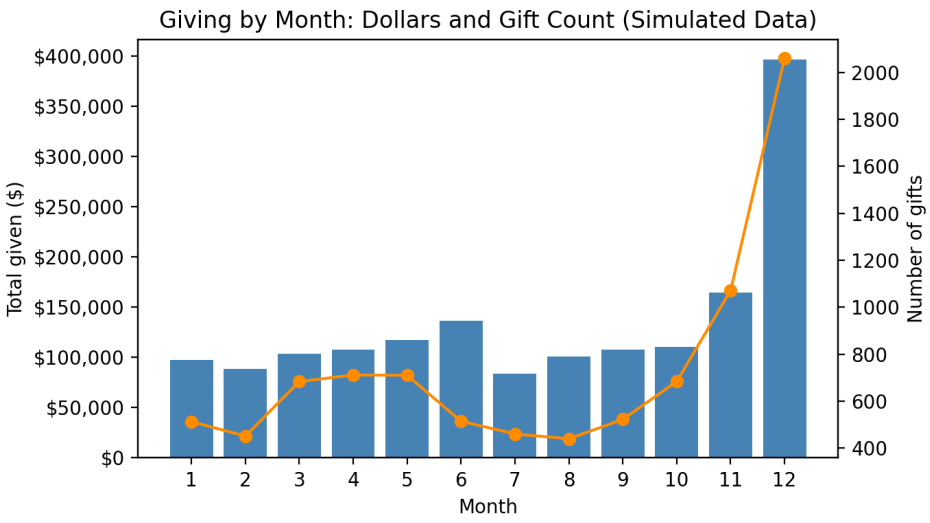


Figure 5. Total giving (bars) and number of gifts (line) by calendar month

What this means

December alone brings in $397,105 (about 25% of all dollars and 23% of gifts, 2,060 gifts). November and December together are about 35% of the year. The spike is driven mostly by the number of gifts, not larger gifts. The small June bump in dollars comes from a few very large major-donor gifts and should be treated as noise.

So what: the year-end window decides the year. Messaging, mailing lists and online giving pages should be ready by autumn.

5.5 Donor type and channel

Channel	Corporate average	Individual average
Event	$403.61	$177.26
Mail	$630.22	$165.40
Online	$496.74	$145.88
Phone	$360.30	$145.68

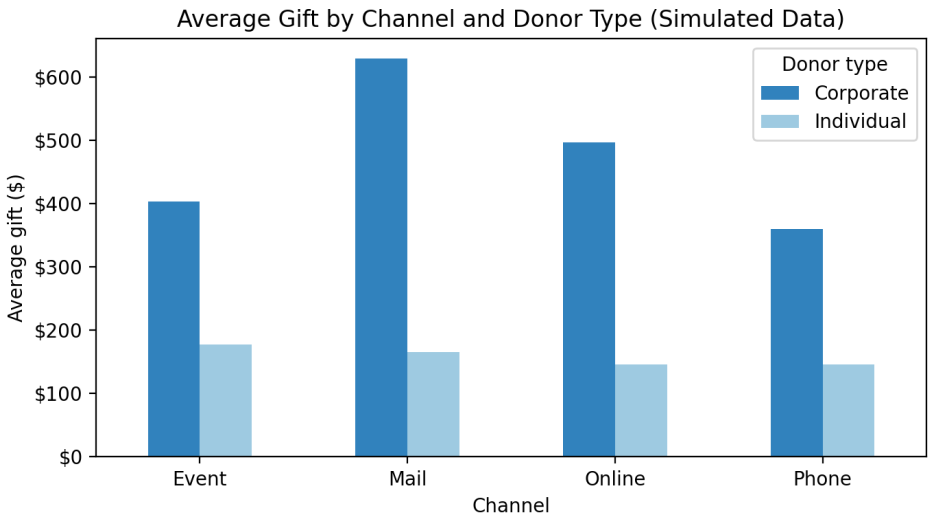


Figure 6. Average gift by channel and donor type

What this means

Individual averages are steady at about $146 to $177 in every channel, as expected because the simulation assigns channel at random. Corporate gifts are 2 to 4 times larger, which fits the built-in 3x multiplier. The corporate differences between channels should not be treated as real: corporate donors are only 8% of donors, so each channel has few gifts and a few large ones swing the average.

6. Predicting Lapse Risk

The analysis above describes what happened. This section asks a forward-looking question: **among donors who gave in 2024, who is likely not to give in 2025?** A model that answers it turns the descriptive findings into a ranked outreach list.

6.1 Design

- **Population:** the 2,230 donors who gave in 2024.
- **Target:** lapsed = no gift in 2025 (1,405 donors, 63.0%).
- **Features:** gift count, total given, average gift, gifts in 2024, months since last gift, a new-in-2024 flag and a corporate flag. All are computed from gifts dated on or before 2024-12-31, so no 2025 information leaks into the model.
- **Evaluation:** a stratified 75/25 split (1,672 training donors, 558 test donors). Two models were compared: logistic regression (an interpretable baseline) and a random forest.

6.2 Results

Model	ROC-AUC (test set)
Logistic regression	0.815
Random forest	0.810
Random guess	0.500

ROC-AUC is the chance that the model ranks a randomly chosen lapsed donor as riskier than a randomly chosen returning donor. A value of 0.5 is a coin flip and 1.0 is perfect.

The model's strength is ranking. When test donors are sorted by predicted risk and split into ten equal groups, the actual lapse rate is about 93% to 95% in the three highest-risk groups and 16% in the lowest, against 63% overall.

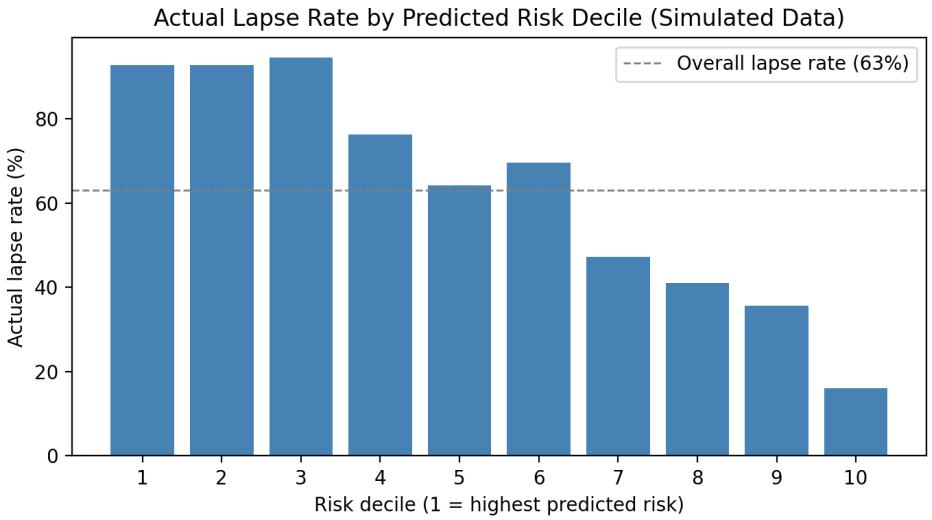


Figure 7. Actual lapse rate by predicted risk decile (test set)

What this means

At a fixed 0.5 threshold, the logistic model is 75% accurate against a 63.1% baseline (always predicting lapse), with 0.90 recall and 0.75 precision on lapsed donors. Accuracy at one threshold is a modest measure here, so ranking quality is the main evidence. In practice the threshold would be set by outreach cost.
The most important features are total given (0.36), average gift (0.25) and gift count (0.14). These are the same signals that separate the hidden donor segments, so the model confirms the patterns built into the simulation. It is not a new discovery.

6.3 How it would be used

Score every current donor, then rank outreach by predicted risk and past giving: valuable donors who are likely to leave come first. This analysis is a backtest, since the 2025 outcome is already known. In production the model would score donors whose outcome is still unknown, and results would be checked with a time-based split on real data.

7. Recommendations

#	Recommendation	Evidence	How to measure success
1	Improve the first renewal	Only about 30% of new donors give again	Second-gift rate within 12 months, and thank-you response rate
2	Steward loyal and major donors	Survivors give 1.7 to 2 times more per gift	Retention rate of loyal and major donors
3	Tier the win-back effort	77 of 1,405 lapsed donors hold half of lapsed giving	Reactivation rate and dollars recovered per tier, per outreach hour
4	Rank outreach by risk and value	ROC-AUC 0.82, with 93% lapse in the top risk groups and 16% in the lowest	Reactivation lift for model-ranked outreach against a control group
5	Plan the Year-End Appeal early	Nov and Dec are about 35% of giving, and the appeal raised $473,775	Year-end revenue against the prior year, and share of gifts made online
6	Keep acquiring, and track cohorts	62% of 2025 giving came from returning donors, and older cohorts shrink every year	Cohort retention reported every year

8. Extension: Salesforce CRM Build

Fundraising teams track donors and gifts in a CRM. To show the same data working in that setting, part of the dataset was rebuilt in a free Salesforce Developer Edition org.

Source	Salesforce object	Notes
Donor (one row in donors.csv)	Account	One Account per donor
Gift (one row in gifts.csv)	Opportunity	Stage set to Closed Won
Donor ID on each gift	Account relationship	Links each gift to its donor through an external ID

- **Sample:** 500 donors (random sample, seed 42) with all of their 901 gifts. The free org has 5 MB of storage, about 2,500 records, and the full data is 13,816 records.

- **Custom fields:** Donor ID (external ID, unique), Donor Type and Donor State on Account. Gift ID (external ID, unique), Gift Campaign and Channel on Opportunity.
- **Import:** the Data Import Wizard loaded the 500 donors (0 failed). dataloader.io loaded the 901 gifts through a lookup on the Donor ID external ID (0 errors). Donors loaded first so every gift had a donor to attach to.
- **Reporting:** three reports (giving by campaign, by month and by donor type) and one dashboard, Donor Giving Overview.
- **Validation:** every report totals 901 records and $150,155, matching the file loaded, and an opened gift record showed the correct donor link.

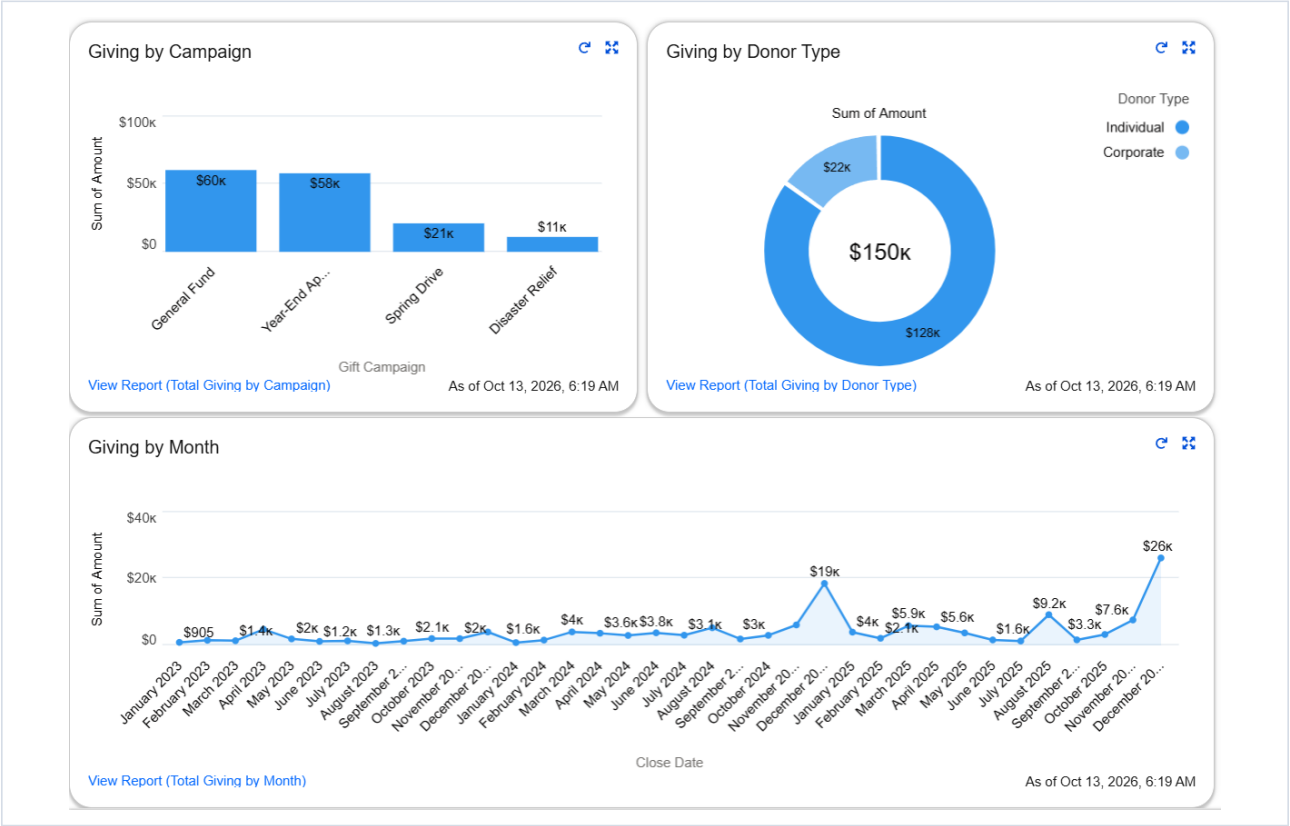


Figure 8. Donor Giving Overview dashboard in Salesforce (campaign, donor type and monthly giving)

Reading the sample

Because Salesforce holds a sample, its totals differ from the full dataset ($1,616,405). For example, General Fund is 40% of the sample but 54% of the full data, since a few large gifts swing a small sample. This extension demonstrates the CRM workflow. It does not produce new findings about donor behavior.

9. Validation

Each key number was calculated in both Python and PostgreSQL. All matched.

Metric	Python	SQL	Match
Row counts (donors, gifts)	5,000 and 8,816	5,000 and 8,816	Yes
Retention 2023 to 2024	29.4%	29.4% (579 of 1,971)	Yes
Retention 2024 to 2025	37.0%	37.0% (825 of 2,230)	Yes
Cohort retention (2023 years 1 and 2, 2024 year 1)	29.4%, 19.1%, 30.8%	29.4%, 19.1%, 30.8%	Yes
Lapsed donors	1,405	1,405	Yes
Donors for 50% and 80% of lapsed giving	77 and 441	77 and 441	Yes
Giving by cohort and year	Six dollar totals	Same values	Yes
Giving by campaign	$866,315; $473,775; $182,260; $94,055	Same values	Yes

The lapse model's target also reproduces the SQL result: 1,405 lapsed donors.

10. Limitations and Next Steps

Limitations

- **Simulated data.** Retention rates, seasonality and gift sizes come from settings chosen in the notebook, so the findings demonstrate the method and do not describe real donor behavior.
- **Major donors add noise.** A few very large gifts swing monthly dollar totals and corporate averages. Gift counts are a steadier measure.
- **The 2025 cohort has one year of history.** Its retention cannot be measured yet.
- **Lapsed is a snapshot.** A donor who gave in 2024 but not 2025 may return in 2026. The simulation allows this at a lower chance.
- **Calendar-year retention is a simple definition.** A donor who gives in December one year and January two years later counts as lapsed for the year in between.
- **Correlation, not cause.** The analysis describes what happened. It does not prove that any outreach would change behavior, which would need a test.
- **Model backtest.** The model was evaluated on a single split of one year of outcomes.

Next steps with real data

1. **RFM scoring** (recency, frequency, monetary value) as a simple, explainable complement to the model.
2. **Outreach experiments**, such as A/B tests of thank-you timing and message, to measure what actually changes behavior.
3. **Time-based model validation** and additional features, such as campaign response and channel history.
4. **A BI dashboard** (for example Tableau) so fundraising staff can track retention by cohort each year.
5. **Data quality rules** in the CRM, such as validation rules and duplicate checks.

Appendix A: Key SQL Queries

Query 1: year-over-year retention (self-join)

```
WITH yearly_donors AS (
    -- One row per donor per giving year
    SELECT DISTINCT donor_id,
           EXTRACT(YEAR FROM gift_date)::INT AS gift_year
    FROM gifts
),
retention AS (
    -- Match each donor-year to the same donor in the next year
    SELECT cur.gift_year,
           COUNT(DISTINCT cur.donor_id) AS donors_in_year,
           COUNT(DISTINCT nxt.donor_id) AS donors_who_returned
    FROM yearly_donors cur
    LEFT JOIN yearly_donors nxt
        ON cur.donor_id = nxt.donor_id
       AND nxt.gift_year = cur.gift_year + 1
    GROUP BY cur.gift_year
)
SELECT gift_year, donors_in_year, donors_who_returned,
       ROUND(100.0 * donors_who_returned / donors_in_year, 1) AS retention_pct
FROM retention
ORDER BY gift_year;
```

The first CTE reduces gifts to one row per donor per year. The self-join looks for the same donor in the following year, and the LEFT JOIN keeps donors who did not return, so COUNT(DISTINCT nxt.donor_id) counts only those with a match.

Query 2: lapsed giving concentration (window functions)

```
WITH donor_totals AS (
    SELECT donor_id, SUM(amount) AS total_given
    FROM gifts
    GROUP BY donor_id
),
lapsed AS (
    -- Gave in 2024 but has no gift in 2025
    SELECT DISTINCT donor_id
    FROM gifts
    WHERE EXTRACT(YEAR FROM gift_date) = 2024
      AND donor_id NOT IN (
          SELECT donor_id FROM gifts
          WHERE EXTRACT(YEAR FROM gift_date) = 2025
      )
),
ranked AS (
    SELECT l.donor_id, t.total_given,
           ROW_NUMBER() OVER (ORDER BY t.total_given DESC, l.donor_id) AS donor_number,
           100.0 * SUM(t.total_given) OVER (ORDER BY t.total_given DESC, l.donor_id)
                 / SUM(t.total_given) OVER () AS cumulative_pct
    FROM lapsed l
    JOIN donor_totals t ON l.donor_id = t.donor_id
)
SELECT COUNT(*) AS total_lapsed_donors,
       MIN(donor_number) FILTER (WHERE cumulative_pct >= 50) AS donors_to_reach_50pct,
       MIN(donor_number) FILTER (WHERE cumulative_pct >= 80) AS donors_to_reach_80pct,
       ROUND(SUM(total_given), 0) AS total_past_giving_of_lapsed
FROM ranked;
```

Ordering by lifetime giving and taking a running total divided by the grand total gives each donor's cumulative share. MIN with FILTER picks the first donor number where that share crosses 50% or 80%.

Appendix B: Repository Contents

File or folder	Contents
README.md	Project overview, key results and how to run
donor_analysis.ipynb	Data simulation, validation and six charts
lapse_prediction.ipynb	Lapse prediction model and three charts
data/	donors.csv (5,000 rows) and gifts.csv (8,816 rows)
sql/	01_schema_and_data_quality.sql, 02_retention_and_lapse_analysis.sql, 03_cohort_and_campaign_analysis.sql
charts/	PNG charts used in this report
salesforce/	Sample CSVs, sampling script, dashboard screenshot
requirements.txt	Python packages